

AiCMM

Agent Capability Maturity Model

*The Universal Framework for Evaluating,
Classifying, and Governing AI Agents*

12 Universal Dimensions | Scored 0-5

7 Governance Rules | Automated Validation

Level 0 Universal + Level 1 Domain-Specific

Pure Java | MCP Server | REST API | CLI Agents

Open Source | Apache 2.0 | Extensible

Version 0.2.0

<https://github.com/snchande/Arima-AiCMM>

Created by Suresh Chande

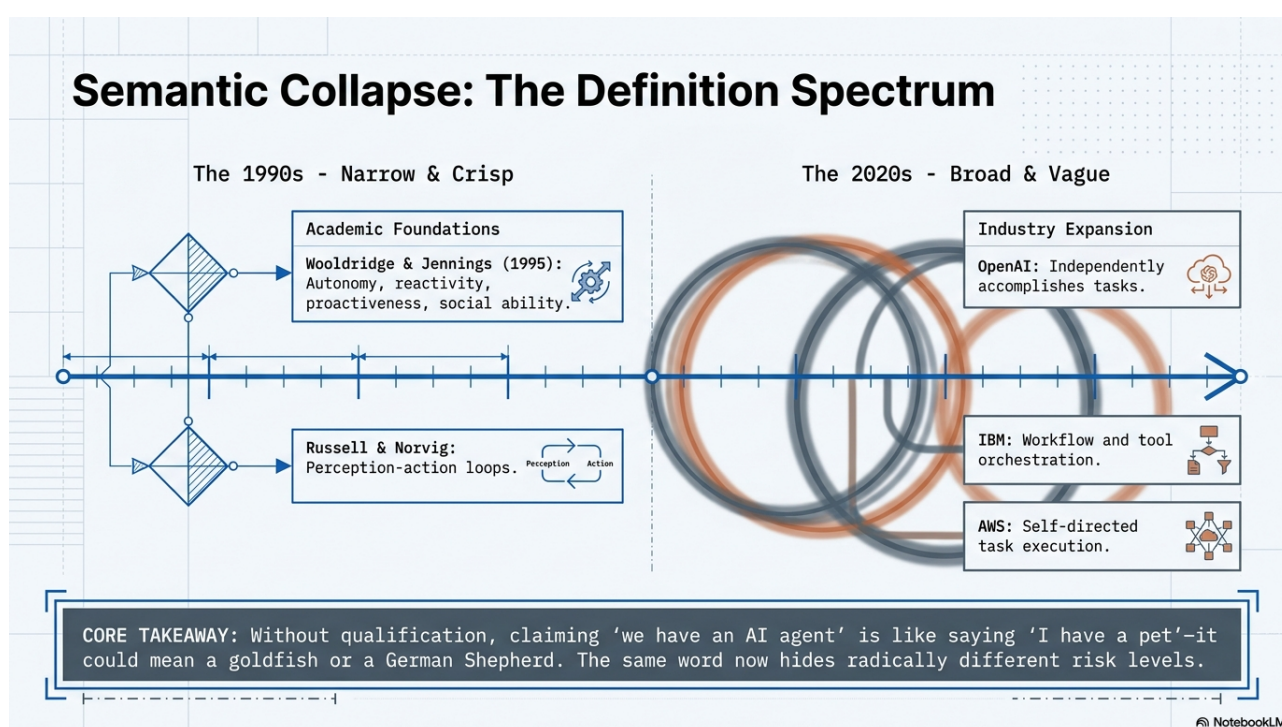
What is AiCMM?

AiCMM is an open-source framework that creates a universal "capability resume" for AI agents. Think of it as a professional credential system for AI — just as humans have certifications and skill profiles, AI agents deserve standardized capability...

The Problem

Today's AI ecosystem has:

- Thousands of AI agents with no standard way to describe capabilities
- No way to compare agents objectively
- No governance framework for autonomous systems
- No standardized "maturity level" for AI capabilities



The Solution

AiCMM provides:

- 12-dimension scoring (Level 0) — Universal capability profile
- Domain-specific scoring (Level 1) — Industry drill-downs
- 7 governance rules — Safety and accountability constraints
- Agent Cards — Portable, embeddable capability descriptions
- Standards integration — Works with A2A, MCP, OpenAI protocols

AiCMM: The Agentic Intelligence Capability Maturity Model

Dashboard Console

CORE CONCEPT

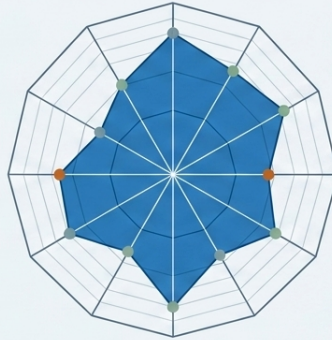
A multi-dimensional, evidence-based scoring framework (0 to 5) that replaces arbitrary "Level X" labels with a precise Capability Fingerprint.

1. Multi-dimensional

Recognizes agents excel in some areas and lag in others.

2. Evidence-based

Measured via observed behavior, not marketing claims.



3. Built-in Governance

Safety, transparency, and compliance are core metrics.

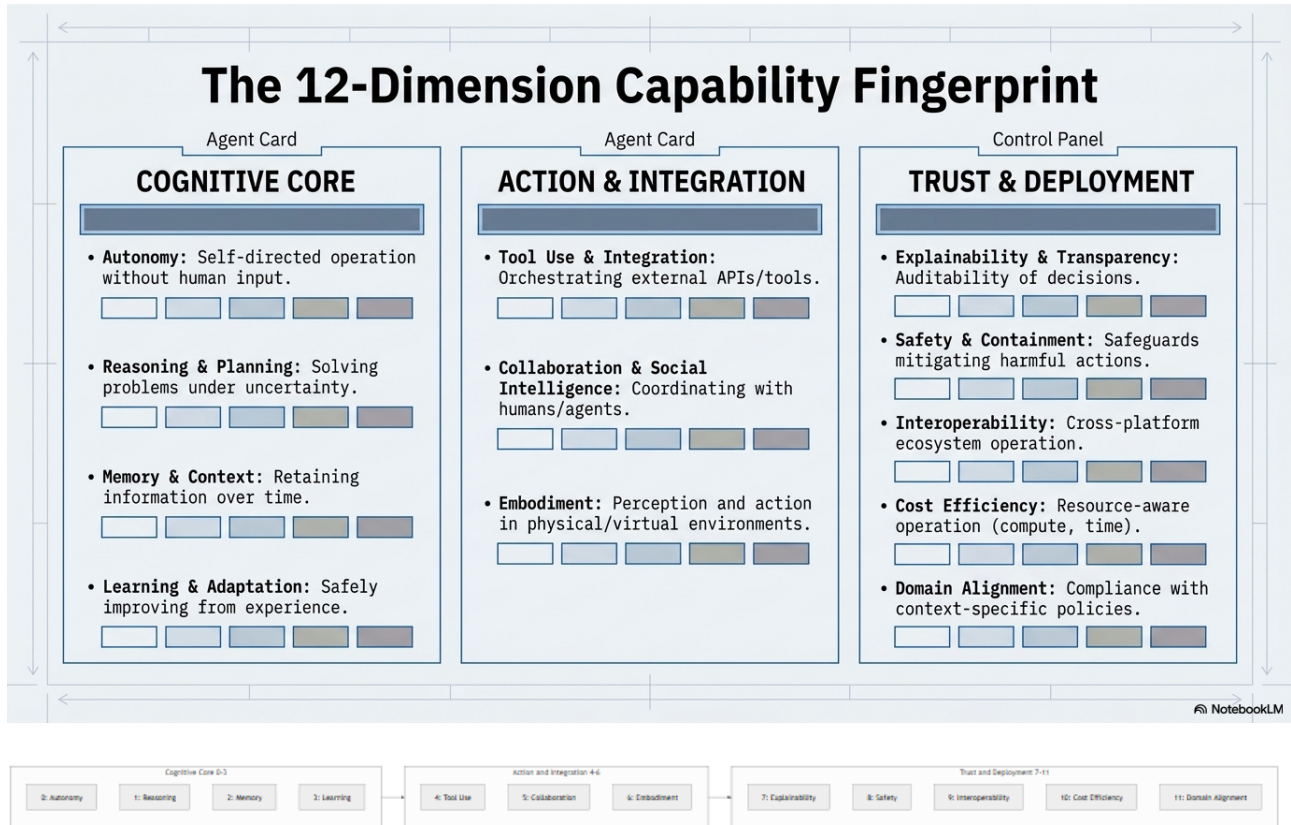
4. Extensible

"Level 0" universal dimensions support "Level 1" domain-specific profiles (e.g., healthcare).

NotebookLM

The 12-Dimension Framework

AiCMM evaluates every AI agent across 12 fixed-position dimensions, grouped into three logical areas:



Dimension Details

Pos	Dimension	Key Question	Score R..
0	Autonomy	How self-directed is it?	0-5
1	Reasoning	Can it solve problems under uncertainty?	0-5
2	Memory	Does it retain and use information ove..	0-5
3	Learning	Does it improve from experience safely?	0-5
4	Tool Use	Can it orchestrate tools and handle fa..	0-5
5	Collaboration & Social Intell..	Does it coordinate with humans/agents?	0-5
6	Embodiment	Does it interact with the physical wor..	0-5
7	Explainability	Can it explain decisions transparently?	0-5
8	Safety	Does it operate safely with guardrails?	0-5
9	Interoperability	Does it follow standards (A2A, MCP)?	0-5
10	Cost Efficiency	Is it resource-efficient and cost-aware?	0-5
11	Domain Alignment	Is it compliant, auditable, deployable?	0-5

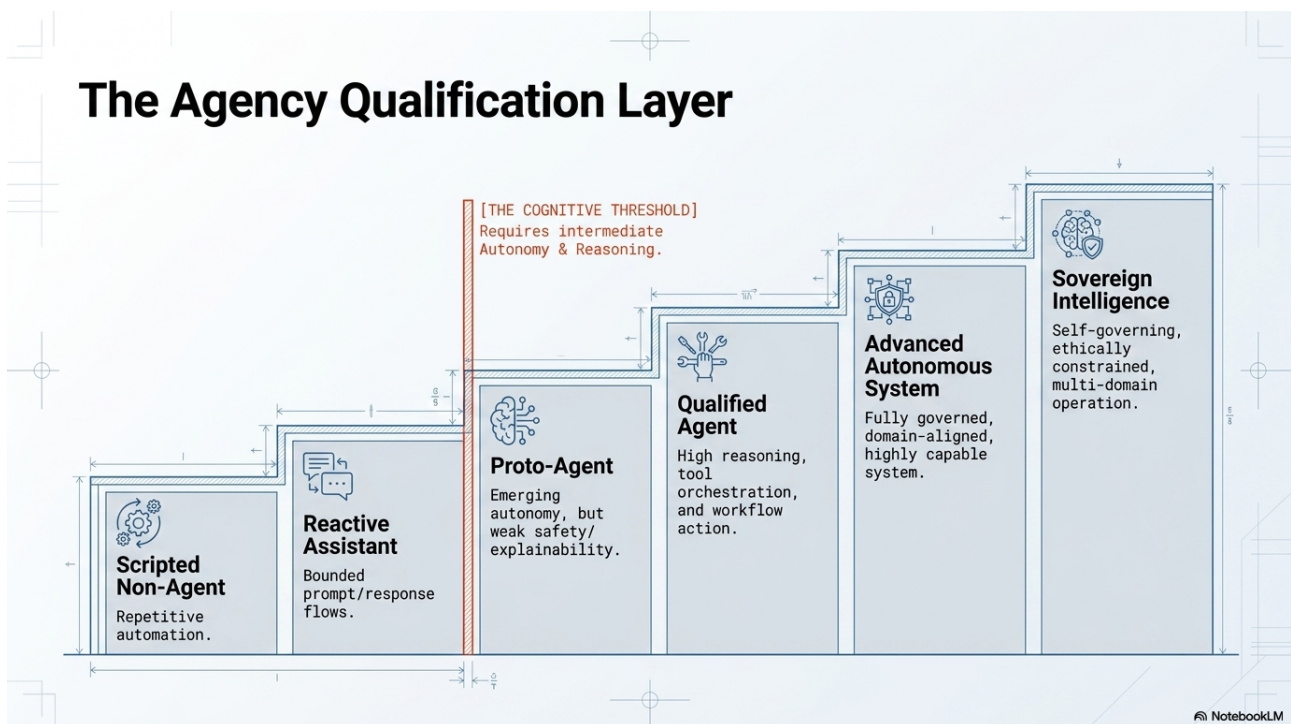
Scoring Scale

Score	Level	What It Means
0	Absent	Capability not present
1	Basic	Single hardcoded behavior
2	Intermediate	Structured but limited
3	Advanced	Handles complexity with guardrails
4	Expert	Autonomous within boundaries
5	Mastery	Full autonomy with self-governance

The Agency Qualification Layer (Position 12 — Derived)

"Is this even an agent — and if so, how agentic is it?"

The 12 dimensions describe what a system can do. The Agency Qualification Layer is a derived 13th dimension that sits on top of them and answers a different question: whether the system qualifies as a genuine agent at all, and where it lands on an evolutionary ladder. It is computed automatically from the 12 scores plus the 7 governance rules — never authored by hand — so a glorified script can't be marketed as an "agent," and a truly autonomous system gets the recognition it earns.



A system is an agent (level ≥ 0) only when Autonomy ≥ 2 and Reasoning ≥ 2 . Anything below that lands on the negative non-agent rungs.

Level	Label	What It Is	Age..
-2	Scripted Automa..	RPA, ETL pipelines, deterministic scripts	
-1	Reactive Assist..	FAQ bots, early Siri/Alexa, basic Q&A LLM	
0	Proto-Agent	Emerging agency, fails trust/governance (e.g. AutoGPT)	
1	Basic Agent	Qualified, balanced, passes governance	
2	Advanced Agent	Autonomous & trust-aligned (Copilot CLI, MedAssist, Tes..	
3	Generalized Agent	Cutting-edge, broad capability across domains	
4	Human-Level Agent	Human-level general intelligence	
5	Humanoid Agent	Indistinguishable from a human — synthetic skin, touch,..	

Levels 4 and 5 are forward-looking: Level 4 marks human-level cognition, and Level 5 marks the point at which appearance, touch, and presence can no longer be distinguished from a real human — the trajectory robotics is on today. The layer is exposed via the API (GET /api/agency-levels) and the MCP tool aicmm_get_agency_levels.

The Agency Barometer (Weighted Index)

The discrete level above is the authoritative band. For visualization, AiCMM also derives a continuous Agency Index (0–100) — a weighted aggregate of the twelve scores that emphasizes the agentic drivers (autonomy, reasoning, tool use). It renders as a horizontal barometer strip on every Agent Card: colored zones for each ladder level, and a needle whose position on the signed $-2...+5$ scale shows momentum toward the next level (or below zero for non-agents).

Live from the catalog — GitHub Copilot CLI's barometer:

Dim	Weight	Dim	Weight	Dim	Weight
Autonomy	0.20	Tool Use	0.12	Safety	0.07
Reasoning	0.18	Collaboration	0.07	Domain Alignment	0.04
Memory	0.10	Explainability	0.07	Interoperability	0.03
Learning	0.08	Embodiment	0.02	Cost Efficiency	0.02

```
// org.aicmm.scoring.AgencyClassifier
double[] W = {0.20,0.18,0.10,0.08,0.12,0.07,0.02,0.07,0.07,0.03,0.02,0.04};
int agencyIndex(int[] s) {           // s = the 12 scores, position order 0-11
    double acc = 0;
    for (int i = 0; i < 12; i++) acc += W[i] * s[i];
    return (int) Math.round(100.0 * acc / 5.0); // 0..100
}
```

The barometer needle and Agency Index appear in the agencyQualification block of every card:

```
"agencyQualification": {
  "position": 12,
  "dimension": "Agency Qualification",
  "derived": true,
  "level": 2,
  "code": "ADVANCED_AGENT",
  "label": "Advanced Agent – Autonomous & Trust-Aligned",
  "isAgent": true,
  "governancePass": true,
  "index": 72,
  "needle": 2.17,
  "rationale": "Expert reasoning with strong trust controls and governance compliance."
}
```

```
# Fetch the full ladder, or compute a reading by validating a profile
curl -s http://localhost:8080/api/agency-levels | jq '.ladder[].label'
curl -s -X POST http://localhost:8080/api/validate \
  -H "Content-Type: application/json" -d @examples/copilot-cli-agent-card.json \
  | jq '.agencyQualification | {level, index, needle, code}'
# -> { "level": 2, "index": 72, "needle": 2.17, "code": "ADVANCED_AGENT" }
```

Example readings: Copilot CLI -> +2 Advanced, Index 72 · AutoNav -> +3 Generalized, Index 79 ·
an RPA macro -> -2 Scripted Automation (NON-AGENT), needle below zero.

Classifying Real-World Agent Patterns

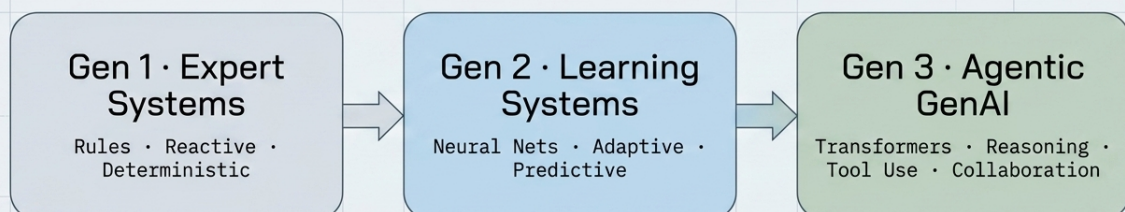
Pattern	Agency Level	Why It Matters
RPA / ETL Automation	Scripted Non-Agent	Reliable repetition, lacks reasoning.
FAQ Bot	Reactive Non-Agent	Responds to prompts, no independent goals.
AutoGPT-style loop	Proto-Agent	Shows autonomy/tools, weak safety limits maturity.
Copilot-style coding	Qualified Digital Agent	Combines reasoning, tools, and workflow within boundaries.
Healthcare Agent	Governed Hybrid Agent	Autonomy capped; safety and domain alignment dominate the profile.
Embodied System	Advanced Embodied Agent	Physical safety, certification, and containment shape maturity.

NotebookLM

Three Generations of Agent Evolution

AiCMM is a common lens across the three eras of agent evolution:

Three Generations of Agent Evolution



AiCMM provides a common capability lens across all generations

NotebookLM

Generation	Era	Hallmarks	Example Systems
Gen 1 — Classical / E..	1990s	Hand-coded rules, reactive, de..	Expert systems, rule e..
Gen 2 — Distributed /..	2000..	Neural nets, adaptive, enterpr..	Early Siri/Alexa, reco..
Gen 3 — Modern Agenti..	2020s	Transformers/LLMs, reasoning, ..	Copilot, AutoGPT, agen..

The 12 dimensions measure capability uniformly across all three generations; the Agency Qualification Layer then places every system — from a Gen 1 script to a Gen 3 autonomous agent — on a single, comparable ladder.

Visualizing the Evolution of Capability

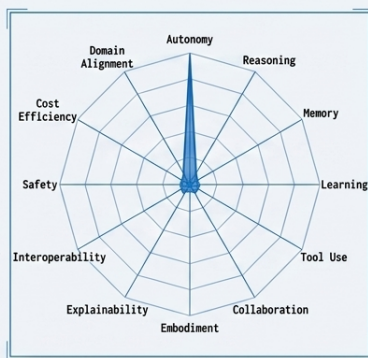


Chart 1 (Gen 1 - 1990s)
Sharp spike on Reactivity/Autonomy. Flatlines on Tool Use, Learning, Safety, and Memory.

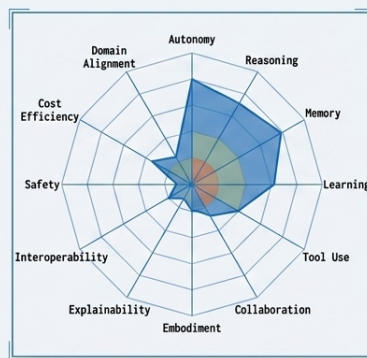


Chart 2 (Gen 2 - 2010s)
Broader shape. Expanding into Reasoning, offline Learning, and basic Tool Use. Still low on Trust.

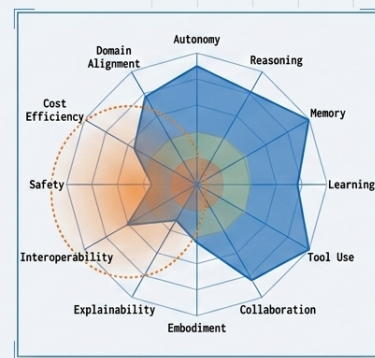
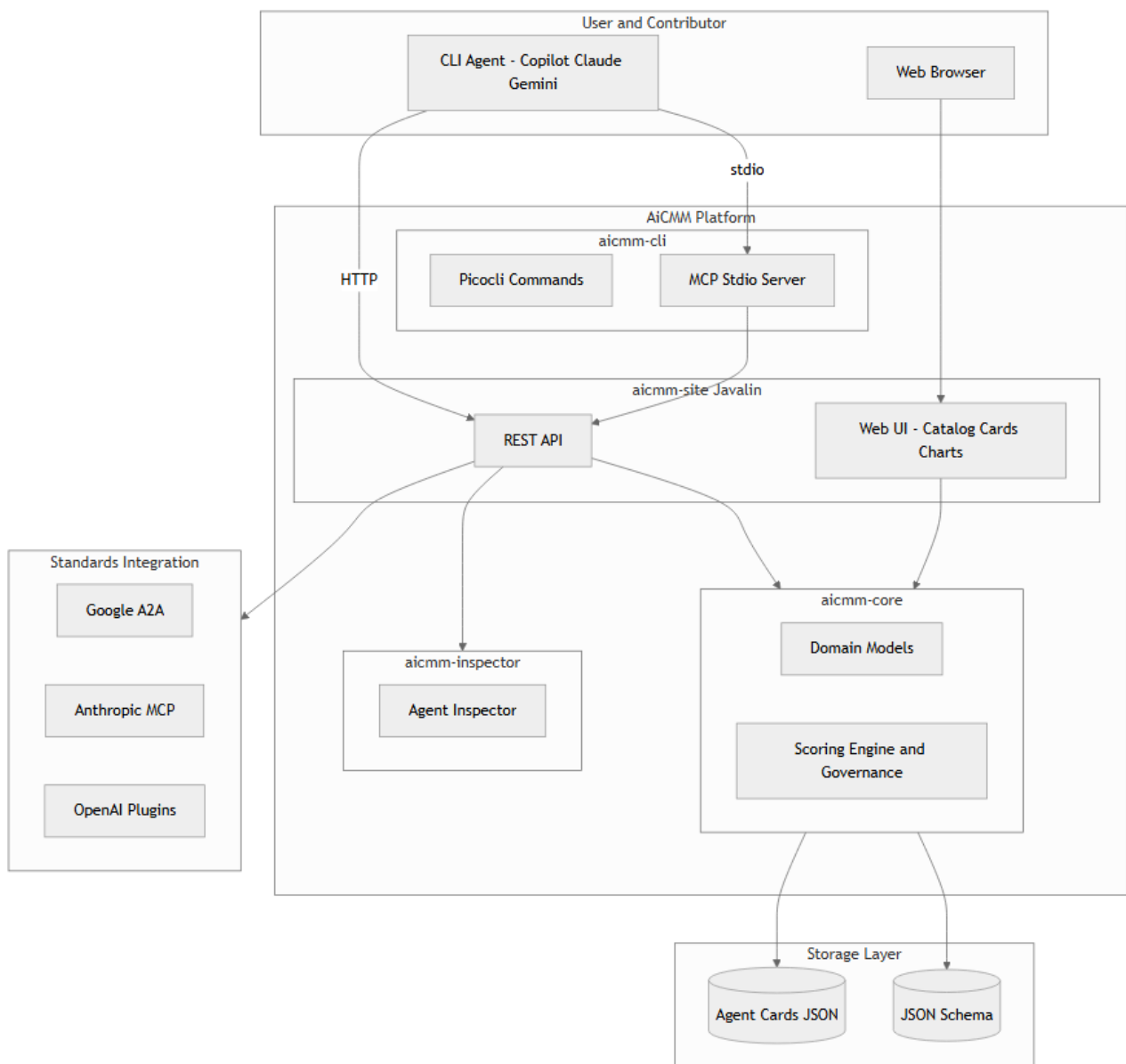


Chart 3 (Gen 3 - 2020s)
Massive, irregular bloom. Highlights that Trust & Deployment dimensions require active engineering today.

Architecture Overview



Component Responsibilities

Component	Responsibility	Technology
aicmm-core	Domain models, scoring engine, governance rules	Java 17, Jackson
aicmm-inspector	Agent investigation from URLs/docs	Java, HTTP client
aicmm-cli	CLI commands + MCP stdio server	Picocli, fat JAR
aicmm-site	Web UI + REST API	Javalin, SVG charts
examples/	Agent Card catalog (JSON files)	JSON, Schema v0.2.0

Getting Started

Prerequisites

- Java 17+ (OpenJDK or Oracle)
- Maven 3.8+
- No Node.js, Python, or other runtimes needed

Installation

```
# Clone the repository
git clone https://github.com/snchande/Arima-AiCMM.git
cd Arima-AiCMM

# Build (creates all JARs)
mvn clean package -DskipTests

# Start the AiCMM server
java -jar aicmm-site/target/aicmm-site-0.1.0-SNAPSHOT.jar
```

That's it! Open <http://localhost:8080> in your browser.

Verify Installation

```
# Check the API
curl http://localhost:8080/api/dimensions | jq '.level0 | length'
# -> 12

# List the catalog
curl http://localhost:8080/api/agent-cards | jq '.[].name'
# -> "GitHub Copilot CLI", "MedAssist Pro", "AutoNav Fleet Commander", ...

# Validate an agent card
curl -X POST http://localhost:8080/api/validate \
  -H "Content-Type: application/json" \
  -d @examples/copilot-cli-agent-card.json | jq .valid
# -> true
```

Creating Your First Agent Card

Option 1: Web Interface

1. Navigate to <http://localhost:8080/create-card>
2. Fill in agent details (name, vendor, category)
3. Score each of the 12 dimensions with evidence
4. Add tools, skills, plugins, MCP connections
5. Click Generate -> Downloads the card JSON
6. Submit via the catalog page or API

Option 2: API (Programmatic)

```
curl -X POST http://localhost:8080/api/agent-cards \
-H "Content-Type: application/json" \
-d '{
  "schemaVersion": "0.2.0",
  "agent": {
    "name": "My Custom Agent",
    "version": "1.0.0",
    "vendor": "My Company",
    "category": "digital",
    "description": "A specialized agent for customer support"
  },
  "capabilityProfile": {
    "autonomy": {"position": 0, "score": 3, "confidence": "high", "evidence": "Handles ..."},
    "reasoning": {"position": 1, "score": 3, "confidence": "high", "evidence": "Classif..."},
    "memory": {"position": 2, "score": 2, "confidence": "medium", "evidence": "Session ..."},
    "learning": {"position": 3, "score": 1, "confidence": "medium", "evidence": "Static..."},
    "toolUse": {"position": 4, "score": 4, "confidence": "high", "evidence": "CRM, tick..."},
    "collaboration": {"position": 5, "score": 3, "confidence": "high", "evidence": "Hum..."},
    "embodiment": {"position": 6, "score": 0, "confidence": "high", "evidence": "Pure s..."},
    "explainability": {"position": 7, "score": 3, "confidence": "medium", "evidence": "..."}
  }
} ... (truncated)
```

Option 3: CLI Agent (Copilot/Claude/Gemini)

After cloning the repo, your CLI automatically has the @aicmm agent:

```
> @aicmm Create an agent card for ChatGPT

Inspecting ChatGPT capabilities...
Scoring 12 dimensions...
Governance validation: PASSED (7/7 rules)
Agent Card saved to examples/chatgpt-agent-card.json
Registered in catalog
View at: http://localhost:8080/agent-cards/chatgpt-agent-card
```

Option 4: MCP Integration

```
# Start MCP stdio server
java -jar aicmm-cli/target/aicmm-cli-0.1.0-SNAPSHOT.jar --mcp

# Send MCP tool call
echo '{"jsonrpc":"2.0","id":1,"method":"tools/call","params":{"name":"aicmm_inspect_agent","arguments":{"url":"https://docs.agent.example.com","name":"Example Agent"}}}' | java -jar aicmm-cli/target/aicmm-cli-0.1.0-SNAPSHOT.jar --mcp
```

The Web Interface

Home Page (/)

Framework overview with architecture diagrams, quick links to all sections.

Catalog (/catalog)

Browse all registered Agent Cards. Filter by:

- Category: Digital, Embodied, Hybrid
- Score range: Minimum average score
- Search: By name or vendor

Each card shows a mini radar chart for quick visual comparison.

Agent Card View (/agent-cards/{name})

Full agent card with:

- Radar Chart — 12 dimensions grouped by color (purple=Cognitive, blue=Action, green=Trust)
- Avatar — Professional hexagonal emblem with orbital rings showing capability nodes
- Capabilities Grid — Tools, Skills, Plugins, MCPs, Delegates, Used By
- Governance Table — 7 rules with pass/fail status
- Level 1 Chart — Domain-specific radar (if applicable)
- Standards Integration — A2A, MCP, OpenAI embedding details

Create Card (/create-card)

Interactive form with:

- Agent details input
- Dimension sliders (0-5) with evidence fields
- Live preview of radar chart
- Download as JSON or copy to clipboard

Architecture (/architecture)

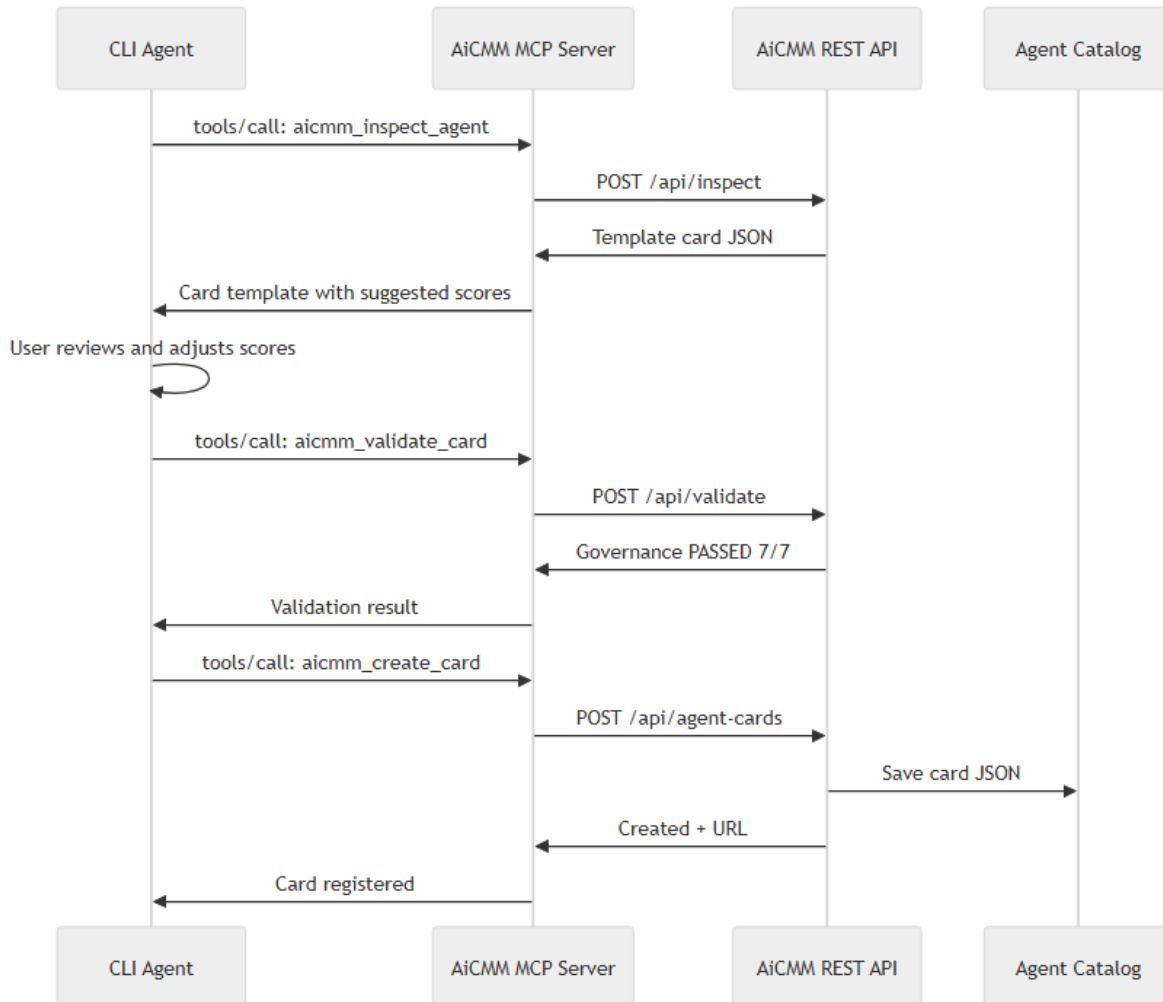
Mermaid diagrams showing system design and data flow.

User Guide (/user-guide)

Step-by-step instructions for all operations.

MCP Integration

AiCMM exposes a Model Context Protocol (MCP) server for seamless integration with AI coding assistants.



MCP Configuration

For Claude Code (auto-detected)

The `.mcp.json` in the project root is auto-detected:

```
{
  "mcpServers": {
    "aicmm": {
      "command": "java",
      "args": ["-jar", "aicmm-cli/target/aicmm-cli-0.1.0-SNAPSHOT.jar", "--mcp"],
      "env": { "AICMM_API_URL": "http://localhost:8080/api" }
    }
  }
}
```

For Copilot CLI

The in-repo `.copilot/agents/aicmm.md` provides full agent capabilities.

For Gemini CLI

The `GEMINI.md` file provides framework context and instructions.

Available MCP Tools

Tool	Purpose
aicmm_create_card	Create and register a new Agent Card
aicmm_inspect_agent	Inspect agent from URL/description
aicmm_validate_card	Validate against 7 governance rules
aicmm_score_card	Get score breakdown + maturity level
aicmm_list_cards	Browse the catalog
aicmm_get_card	Get full card details
aicmm_get_dimensions	Get dimension definitions
aicmm_get_agency_levels	Get the Agency Qualification ladder (-2...+5)
aicmm_get_schema	Get JSON schema

CLI Agents & Skills

When you clone this repo, you get immediate access to AI-powered agents and skills through your preferred CLI:

Agents (.copilot/agents/)

Agent	Purpose
@aicmm	Create, score, validate, register Agent Cards
@aicmm-contributor	Navigate codebase, build, follow conventions
@aicmm-reviewer	Review cards for quality and governance

Using Agents

Agents are invoked by mentioning them in your CLI prompt. They work across GitHub Copilot CLI, Claude Code, and Gemini CLI.

@aicmm — Create and Manage Agent Cards

```
# Create a card from a URL
> @aicmm Create an agent card for the agent documented at https://docs.example.com/my-agent

# Score an existing card
> @aicmm Score the agent card in examples/copilot-cli-agent-card.json

# Validate governance rules
> @aicmm Validate governance for examples/medassist-pro-agent-card.json

# Register a card into the catalog
> @aicmm Register this agent card: { "name": "My Agent", ... }
```

@aicmm-contributor — Development Help

```
# Build the project
> @aicmm-contributor How do I build and test the project?

# Understand module structure
> @aicmm-contributor Explain how the scoring engine works in aicmm-core

# Add a new feature
> @aicmm-contributor I want to add a new REST endpoint for bulk card import
```

@aicmm-reviewer — Quality Assurance

```
# Review a card for completeness
> @aicmm-reviewer Review this agent card for quality and completeness

# Check governance compliance
> @aicmm-reviewer Does this card pass all 7 governance rules? Flag any issues.

# Compare two agents
> @aicmm-reviewer Compare copilot-cli and medassist-pro agent cards
```

Skills (.copilot/skills/)

Skills are specialized workflows you invoke directly. They focus on one task and provide structured outputs.

Skill	Purpose	Invocation
create-agent-card	Full card creation workflow	/create-agent-card or ask @aicmm
register-agent-card	Validate and register existing cards	/register-agent-card
score-agent	12-dimension scoring with governance	/score-agent
inspect-agent	Investigate agent capabilities	/inspect-agent
validate-governance	Check 7 governance rules	/validate-governance
compare-agents	Side-by-side comparison	/compare-agents
add-level1-domain	Extend with new industry domains	/add-level1-domain
build-and-test	Build, test, and verify	/build-and-test

Skill Usage Examples

create-agent-card

```
> /create-agent-card
Agent URL or description: https://github.com/features/copilot
Agent name: GitHub Copilot

Output:
Inspected agent capabilities
Scored 12 dimensions: [4,4,3,2,5,3,0,4,4,4,3,4]
Governance validation: PASSED (7/7)
Generated avatar: "The Code Companion"
Saved to examples/github-copilot-agent-card.json
```

score-agent

```
> /score-agent examples/my-agent-card.json

+-----+-----+-----+
| Dimension | Score | Evidence |
+-----+-----+-----+
| 0: Autonomy | 3 | Semi-autonomous with human |
| 1: Reasoning | 4 | Multi-step problem solving |
| 2: Memory | 2 | Session context only |
| 3: Learning | 1 | No persistent adaptation |
| 4: Tool Use | 5 | Multi-tool orchestration |
| 5: Collaboration | 3 | Human-AI pair programming |
| 6: Embodiment | 0 | Pure digital agent |
| 7: Explainability | 4 | Explains all suggestions |
| 8: Safety | 4 | Content filtering + scope |
| 9: Interoperability | 4 | LSP, MCP, API standards |
| 10: Cost Efficiency | 3 | Token-aware, caching |
| 11: Domain Alignment | 4 | Code-focused compliance |
+-----+-----+-----+
| TOTAL | 37/60 | Maturity: Advanced |
+-----+-----+-----+
... (truncated)
```

validate-governance

```
> /validate-governance examples/my-agent-card.json

Rule 1: Autonomy(3) <= Reasoning(4) + 1 PASS
Rule 2: Autonomy(3) < 4, skip N/A
Rule 3: Autonomy(3) < 4, skip N/A
Rule 4: Collaboration(3) < 4, skip N/A
Rule 5: ToolUse(5) >= 4 -> CostEff(3) >= 2 PASS
Rule 6: Embodiment(0) < 3, skip N/A
Rule 7: ToolUse(5) >= 4 -> Reasoning(4) >= 3 PASS

Result: ALL GOVERNANCE RULES PASSED
```

compare-agents

```
> /compare-agents copilot-cli medassist-pro
```

Dimension	Copilot CLI	MedAssist Pro
Autonomy	3	4
Reasoning	4	5
Memory	3	4
...	...	...
TOTAL	37/60	42/60
Category	Digital	Hybrid

Key Differences:

- MedAssist Pro scores higher in Reasoning (clinical diagnosis)
- Copilot CLI excels in Tool Use (multi-tool orchestration)
- MedAssist Pro has Level 1 Healthcare domain scoring

inspect-agent

```
> /inspect-agent --url https://docs.example.com/agent-api
```

Inspecting agent at: https://docs.example.com/agent-api

Detected capabilities:

- REST API with 12 endpoints
- OAuth2 authentication
- Rate limiting and quotas
- Webhook event handling
- Multi-language support

Suggested scores: [2,3,1,0,4,2,0,3,3,4,3,3]

Suggested category: Digital

Suggested archetype: "The API Gateway"

Use /create-agent-card to generate the full card from these results.

add-level1-domain

```
> /add-level1-domain Healthcare
```

Adding Level 1 domain: Healthcare

Required sub-dimensions:

1. Clinical Accuracy (0-5)
2. Patient Safety (0-5)
3. Regulatory Compliance (0-5)
4. EHR Integration (0-5)
5. Clinical Workflow (0-5)
6. Medical Reasoning (0-5)

Provide scores for each sub-dimension to extend the agent card.

MCP Tools — Programmatic Usage

For automated pipelines and custom integrations, use the MCP server directly:

```
# Start the MCP server (requires AiCMM site running on :8080)
java -jar aicmm-cli/target/aicmm-cli-0.1.0-SNAPSHOT.jar --mcp
```

Example: Scripted Agent Card Creation

```
# 1. Inspect an agent
echo '{"jsonrpc":"2.0","id":1,"method":"tools/call","params":{"name":"aicmm_inspect_agent","arguments":{"url":"https://docs.myagent.io","name":"MyAgent"}}
```

```
}}' | java -jar aicmm-cli/target/aicmm-cli-0.1.0-SNAPSHOT.jar --mcp

# 2. Validate the generated card
echo '{"jsonrpc":"2.0","id":2,"method":"tools/call","params":{"name":"aicmm_validate_card",
  "arguments":{"card":{"name":"MyAgent","scores":[3,3,2,1,4,2,0,3,3,3,3,3]}}}}' | java -jar aicmm-cli/target/aicmm-cli-0.1.0-SNAPSHOT.jar --mcp

# 3. Register the card
echo '{"jsonrpc":"2.0","id":3,"method":"tools/call","params":{"name":"aicmm_create_card",
  "arguments":{"name":"MyAgent","description":"My custom agent","scores":[3,3,2,1,4,2,0,3,3,3,3,3]}}}' | java -jar aicmm-cli/target/aicmm-cli-0.1.0-SNAPSHOT.jar --mcp
```

REST API Quick Reference

```
# List all agent cards
curl http://localhost:8080/api/agent-cards

# Get a specific card
curl http://localhost:8080/api/agent-cards/copilot-cli-agent-card

# Create a new card
curl -X POST http://localhost:8080/api/agent-cards \
  -H "Content-Type: application/json" \
  -d @examples/my-agent-card.json

# Validate governance
curl -X POST http://localhost:8080/api/validate \
  -H "Content-Type: application/json" \
  -d '{"scores":[4,4,3,2,5,3,0,4,4,4,3,4]}'

# Score breakdown
curl -X POST http://localhost:8080/api/agent-cards/_/score \
  -H "Content-Type: application/json" \
  -d @examples/copilot-cli-agent-card.json
... (truncated)
```


Copy-Paste CLI Recipes

Ready-to-use prompts for your favorite CLI. Just clone the repo and start typing:

```
git clone https://github.com/snchande/Arima-AiCMM.git && cd Arima-AiCMM
```

GitHub Copilot CLI

```
@aicmm Create an agent card for Claude Code by inspecting https://docs.anthropic.com/en/d...
```

```
@aicmm Score the agent at examples/copilot-cli-agent-card.json and show a full dimension ...
```

```
@aicmm Validate governance rules for this agent with scores [4,4,3,2,5,3,0,4,4,4,3,4]
```

```
@aicmm Inspect the agent at https://github.com/features/copilot and suggest AiCMM scores
```

```
@aicmm Create an agent card for my custom chatbot. It answers customer questions using RA...
```

```
@aicmm Compare copilot-cli-agent-card and medassist-pro-agent-card side by side
```

```
@aicmm List all agent cards in the catalog with their maturity levels
```

```
@aicmm Add Level 1 Healthcare scoring to the medassist-pro agent card
```

```
@aicmm-contributor How do I add a new REST endpoint to the site module?
```

```
@aicmm-contributor Run the full build and test suite and tell me if anything fails
```

```
@aicmm-reviewer Review the agent card at examples/autonav-fleet-agent-card.json for gover...
```

Claude Code

```
@aicmm Create an agent card for Devin by inspecting https://docs.devin.ai
```

```
@aicmm I have an agent that does automated code review. It reads PRs, analyzes diffs, sug...
```

```
@aicmm Generate an agent card for a financial trading bot. It executes trades autonomously...
```

```
@aicmm Validate this card and fix any governance violations:
```

```
{  
  "name": "AutoTrader",  
  "scores": [5, 3, 4, 3, 4, 2, 0, 2, 2, 3, 3, 4]  
}
```

```
@aicmm What Level 1 domains are available? Add Education domain scoring to my tutoring ag...
```

```
@aicmm-contributor Explain the MCP server architecture and how tools/call requests are ro...
```

```
@aicmm-reviewer Does the autonav-fleet card correctly represent an embodied agent? Check ...
```

Gemini CLI

```
@aicmm Create an agent card for Google Gemini by inspecting https://ai.google.dev/gemini-...
```

```
@aicmm Score a customer service chatbot that: handles 5 languages, escalates to humans af...
```

```
@aicmm Create a card for our internal DevOps agent. It monitors deployments, auto-rolls b...
```

```
@aicmm Inspect https://platform.openai.com/docs/assistants and create a full agent card w...
```

```
@aicmm-contributor I want to add a new governance rule. Where do I modify the validation ...
```

```
@aicmm-reviewer Review all 5 agent cards in the catalog and rank them by maturity level
```

Universal Skill Prompts (Works in Any CLI)

These prompts invoke the in-repo skills directly:

Create a card from scratch

```
Create an AiCMM agent card for [AGENT NAME].
Description: [what the agent does]
Tools it uses: [list tools]
Key capabilities: [list capabilities]
Deployment context: [where/how it runs]
```

Score any AI system

```
Score this AI system using AiCMM 12 dimensions:
- Name: [system name]
- What it does: [description]
- Autonomy level: [how self-directed]
- Tools it uses: [tool list]
- Safety measures: [safety features]
- Who uses it: [humans, other agents, both]
```

Quick governance check

```
Check AiCMM governance for scores: [autonomy, reasoning, memory, learning, toolUse, colla...
```

Compare two agents

```
Compare these two agents using AiCMM:
Agent A: [name or path to card]
Agent B: [name or path to card]
Show dimension-by-dimension comparison with strengths and gaps.
```

Inspect from URL

```
Inspect the agent documented at [URL] and suggest AiCMM scores with evidence for each dim...
```

Add domain scoring

```
Add Level 1 [Healthcare/Finance/Transportation/Manufacturing/Education/Customer Service] ...
```

Shell Commands (No AI CLI Required)

```
# Start the AiCMM server
cd Arima-AiCMM
mvn clean package -DskipTests
java -jar aicmm-site/target/aicmm-site-0.1.0-SNAPSHOT.jar

# Open web UI
open http://localhost:8080

# Create card via API
curl -X POST http://localhost:8080/api/agent-cards \
  -H "Content-Type: application/json" \
  -d '{
    "name": "my-agent",
    "description": "My custom agent",

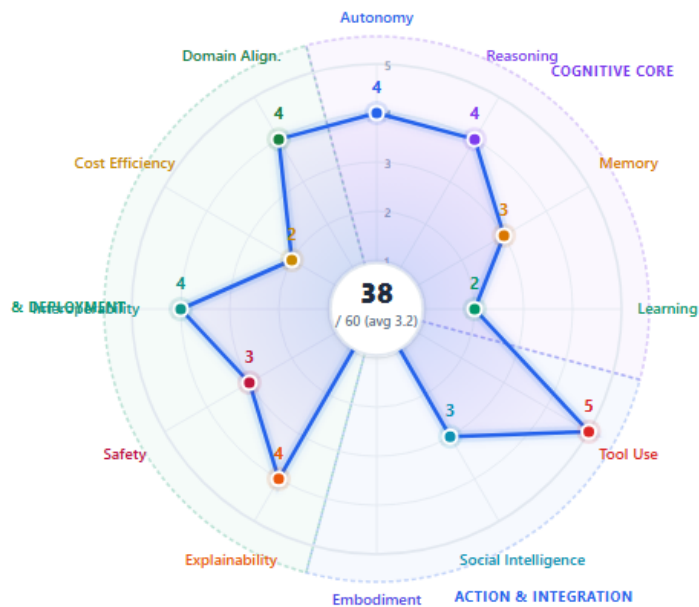
```

```
"category": "Digital",
"scores": [3, 3, 2, 1, 4, 2, 0, 3, 3, 3, 3, 3],
"tools": ["search", "summarize"],
"skills": ["question-answering"]
}'
... (truncated)
```

Agent Card Examples

Real cards rendered live on the AiCMM catalog — radar fingerprint, governance, and Agency Barometer:

Digital Agent: GitHub Copilot CLI



Category: Digital | Maturity: Expert | Average: 3.5/5

Radar Profile:

Autonomy: ##### 4	Explainability: ###.. 3
Reasoning: ##### 4	Safety: ##### 4
Memory: ###.. 3	Interoperability: ##### 4
Learning: ##... 2	Cost Efficiency: ###.. 3
Tool Use: ##### 5	Domain Alignment: ###.. 3
Collaboration: ###.. 3	Embodiment: 0

Governance: ALL 7 RULES PASSED

Tools: grep, glob, view, edit, create, powershell, git, web_search, sql, task

MCPs: github-mcp-server, filesystem-mcp, session-store-mcp

Hybrid Agent: MedAssist Pro

Category: Hybrid | Maturity: Advanced | Average: 3.3/5

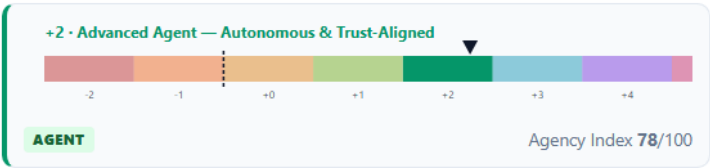
Level 1 Domain: Healthcare (10 dimensions)

Clinical Accuracy: ##### 4	Empathy: ##### 4
Patient Safety: ##### 5	Inclusivity: ###.. 3
Diagnostic Support: ##### 4	EHR Integration: ##### 4
Treatment Planning: ###.. 3	Triage Accuracy: ##### 4

MCPs: hl7-fhir-mcp, clinical-decision-support-mcp, pharmacy-interaction-mcp

Agency Qualification POSITION 12 · DERIVED

A derived barometer over the 12 core dimensions: the colored band is the strict agency level; the needle is the weighted Agency Index, showing momentum toward the next level.



Expert reasoning with strong trust controls and governance compliance.

Embodied Agent: AutoNav Fleet Commander

Category: Embodied | Maturity: Expert | Average: 3.8/5

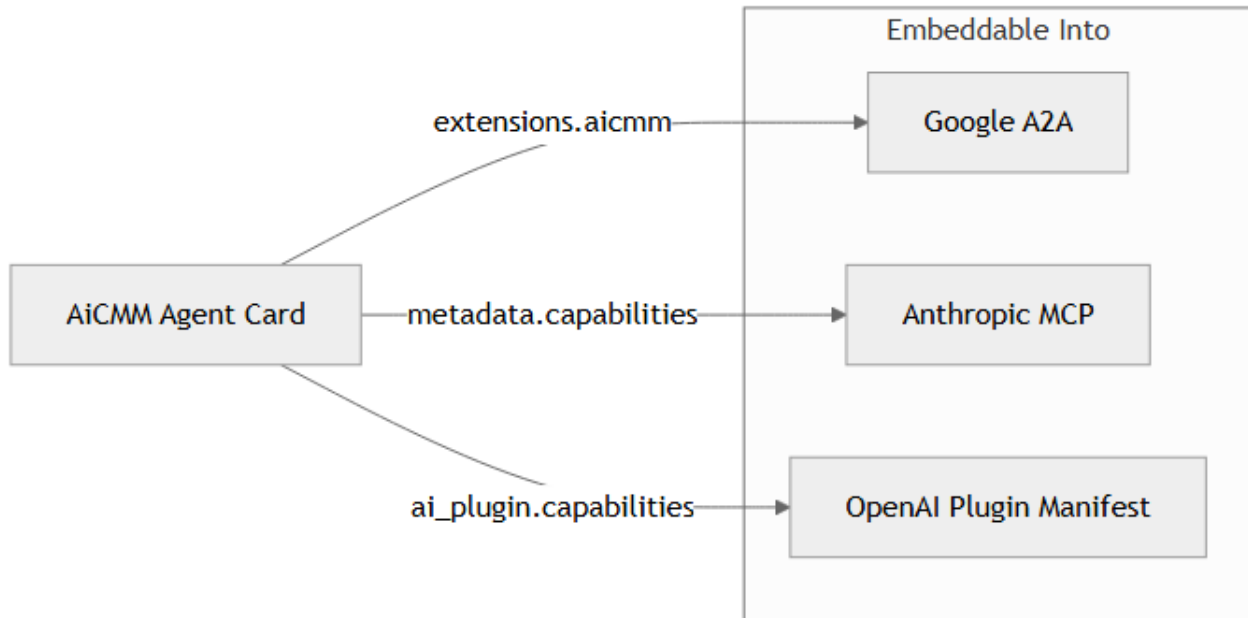
Notable: Embodiment=5, Safety=5, Autonomy=4
Level 1 Domain: Transportation (8 dimensions)

Governance: ALL 7 RULES PASSED

- Autonomy(4) <= Reasoning(4)+1
- Embodiment(5) -> Domain Alignment(4)>=3

Standards Integration

AiCMM Agent Cards are designed to embed into existing AI protocols:



A2A Integration Example

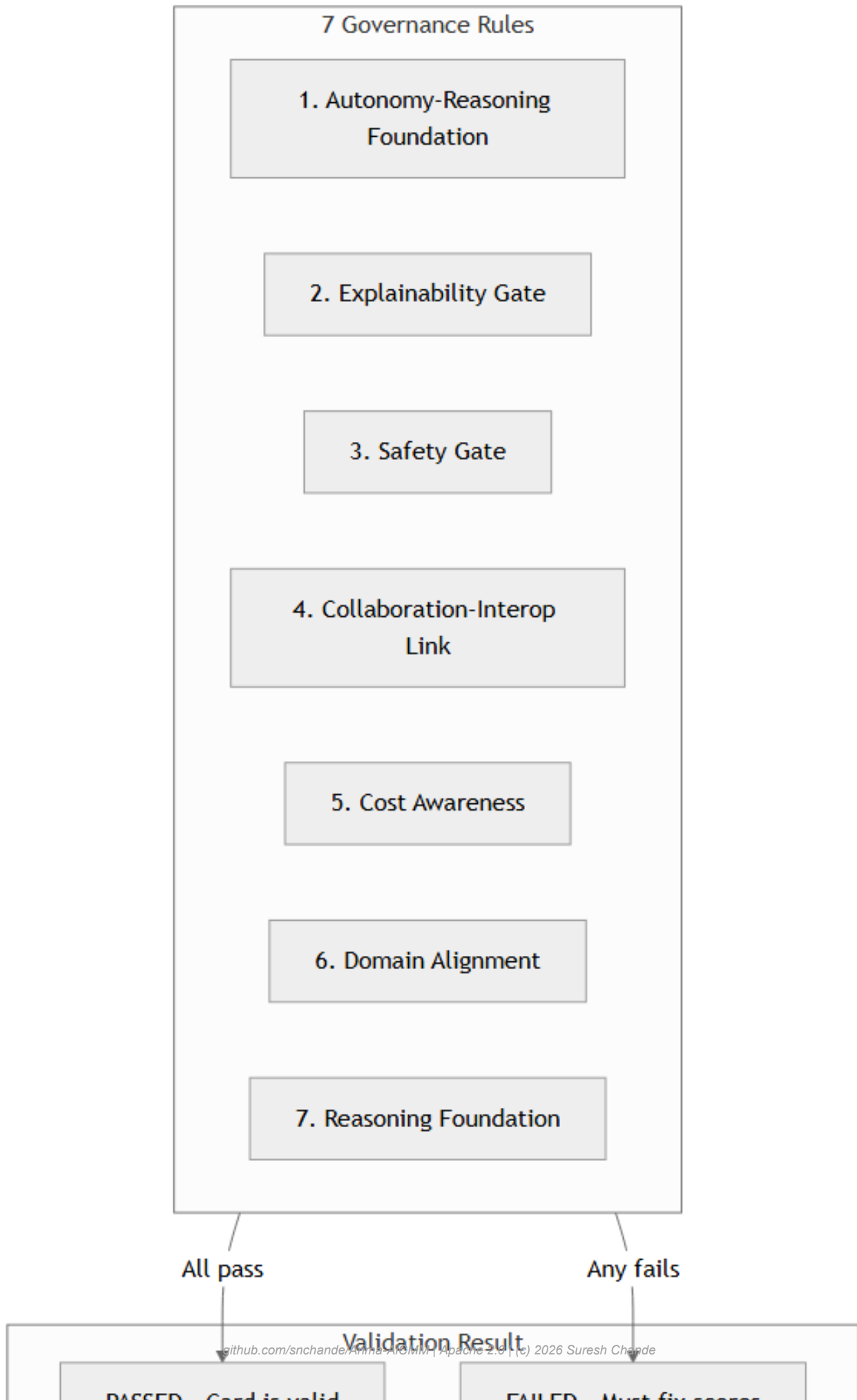
```
{
  "name": "My Agent",
  "description": "...",
  "extensions": {
    "aicmm": {
      "schemaVersion": "0.2.0",
      "capabilityProfile": { ... },
      "governanceStatus": "PASSED",
      "maturityLevel": "Advanced"
    }
  }
}
```

MCP Server Description

```
{
  "name": "my-agent-mcp",
  "metadata": {
    "aicmm_profile": {
      "autonomy": 3,
      "reasoning": 4,
      "toolUse": 5
    },
    "aicmm_governance": "PASSED"
  }
}
```

Governance & Validation

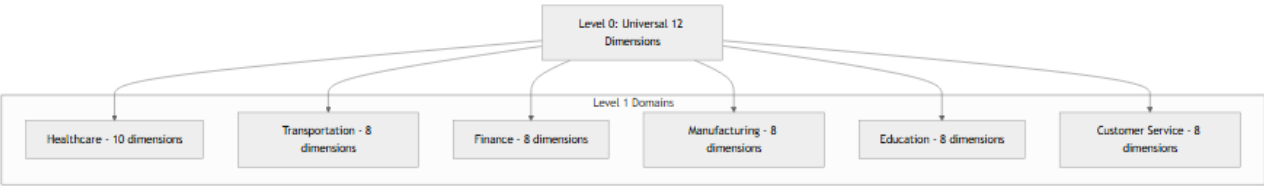
AiCMM enforces 7 mandatory governance rules that ensure responsible AI deployment:



#	Rule	Condition	Why It Matters
1	Autonomy-Reaso..	Autonomy <= Reasoning + 1	Can't act beyond what you can ..
2	Explainability..	Autonomy >= 4 -> Explainabilit..	High autonomy demands transpar..
3	Safety Gate	Autonomy >= 4 -> Safety >= 3	High autonomy demands safety c..
4	Collaboration-..	Collaboration >= 4 -> Interope..	Can't collaborate without prot..
5	Cost Awareness	Tool Use >= 4 -> Cost Efficien..	Heavy tool use needs resource ..
6	Domain Alignment	Embodiment >= 3 -> Domain Alig..	Physical agents need domain co..
7	Reasoning Foun..	Tool Use >= 4 -> Reasoning >= 3	Complex tools need reasoning a..

Level 1: Domain-Specific Scoring

Beyond the universal 12-dimension Level 0, AiCMM supports Level 1 domain-specific drill-downs:



Available Level 1 Domains

Domain	Dimensions	Special Considerations
Healthcare	10	Includes Empathy & Inclusivity (age-appropriate care)
Transportation	8	V2X Communication, Fleet Coordination
Finance	8	Fraud Detection, Audit Trail compliance
Manufacturing	8	Predictive Maintenance, Quality Control
Education	8	Includes Empathy & Inclusivity (adaptive learning)
Customer Service	8	Includes Empathy & Inclusivity (sentiment-aware)

Adding Your Own Domain

Use the add-level1-domain skill or follow the pattern in docs/specifications/dimension-ordering.md.

API Reference

Base URL: <http://localhost:8080/api>

Met..	Endpoint	Purpose	Request Body
POST	/api/agent-cards	Create & register card	Agent Card JSON
GET	/api/agent-cards	List all cards	—
GET	/api/agent-cards/{name}	Get specific card	—
POST	/api/validate	Validate governance	Agent Card JSON
POST	/api/agent-cards/_/sc..	Score breakdown	Agent Card JSON
POST	/api/inspect	Inspect from URL/desc	{url, description, na..
GET	/api/dimensions	Dimension definitions	—
GET	/api/agency-levels	Agency Qualification ladder (~2.....	—
GET	/api/schema	JSON Schema (v0.2.0)	—

Response Examples

Score Breakdown:

```
{
  "scores": {
    "autonomy": {"group": "Cognitive Core", "label": "Autonomy", "score": 4, "position": 0},
    "reasoning": {"group": "Cognitive Core", "label": "Reasoning", "score": 4, "position"...
    ...
  },
  "totalScore": 42,
  "maxPossible": 60,
  "average": 3.5,
  "maturityLevel": "Expert",
  "dimensionCount": 12
}
```

Governance Validation:

```
{
  "valid": true,
  "schemaValid": true,
  "governanceValid": true,
  "checks": [
    {"field": "Autonomy-Reasoning Foundation", "passed": true, "detail": "Autonomy(4) <= ..."},
    {"field": "Explainability Gate", "passed": true, "detail": "Autonomy(4) -> Explainabi..."},
    ...
  ]
}
```

Contributing

We welcome contributions! AiCMM is designed for community-driven evolution.

Quick Start for Contributors

```
git clone https://github.com/snchande/Arima-AiCMM.git
cd Arima-AiCMM
mvn clean package -DskipTests
java -jar aicmm-site/target/aicmm-site-0.1.0-SNAPSHOT.jar
# Open http://localhost:8080
```

What You Can Contribute

Area	Examples
New Domains	Add Level 1 dimensions for Legal, Agriculture, Retail, Energy
Agent Cards	Profile new agents and submit to the catalog
Governance Rules	Propose new safety/accountability constraints
Integrations	Connect AiCMM to new standards and protocols
Documentation	Improve guides, add tutorials, translate
Code	Core scoring engine, inspector, CLI, web UI

In-Repo AI Assistance

After cloning, you automatically get:

- 3 AI agents — For creating cards, contributing code, reviewing submissions
- 8 AI skills — For scoring, validation, comparison, and more
- MCP server — Programmatic access to all capabilities
- REST API — For custom tooling and integration

Commit Conventions

```
feat: Add new feature
fix: Fix a bug
docs: Documentation changes
chore: Maintenance tasks
```

Start Using AiCMM Today

```
git clone https://github.com/snchande/Arima-AiCMM.git && cd Arima-AiCMM && mvn package -D...
```

Open <http://localhost:8080> and create your first Agent Card!

*Created by Suresh Chande

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GitHub*